

The Ohio State University

Department of Computer Science and Engineering

Quality-Stratified Low-Rank Adaptation

for Fine-Tuning Large Language Models on Private Document Collections

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Organizations Need LLMs Grounded in Their Own Documents

The Challenge

- **Critical Knowledge:** Enterprise knowledge bases, legal archives, manuals, and records contain critical organizational knowledge.
- **Strict Privacy:** These documents are private and completely absent from public LLM pre training datasets.
- **Sensitivity:** Data is far too sensitive to transmit to external API providers.
- **On-Premise Need:** Organizations require secure, local models with zero data exposure.

Organizations Need LLMs Grounded in Their Own Documents

searchOption A: Retrieval-Augmented Generation (RAG)

- Best suited when documents are frequently updated at inference time.
- Requires maintaining a real-time vector index, introducing retrieval latency.
- Prone to failure when retrieval misses the critical relevant passages.

psychologyOption B: Fine-Tuning (This Thesis)

- Deep Knowledge: Core domain facts are baked directly into model weights.
- 100% On-Premise: Operates fully locally with no external APIs.
- Low Latency: Fast inference speed with secure deployment.
- Better Fit: Well-suited for low-latency, air gapped, or proprietary corpora.

Why Parameter-Efficient Fine-Tuning?

Full Fine-Tuning

~100%

of parameters updated

- Adam optimizer state alone needs **~2N extra** parameters on top of weights and gradients
- Often exceeds single-GPU memory at billions of scale, needs model parallelism
- Risk of catastrophic forgetting, can overwrite pretrained knowledge

PEFT / LoRA (This Thesis)

<1%

of parameters trained (adapters only)

- Base weights frozen, only small low-rank adapters are trained
- Memory dominated by the frozen base model, not gradients + optimizer state
- Lower risk of forgetting; adapters merge at inference for zero added latency

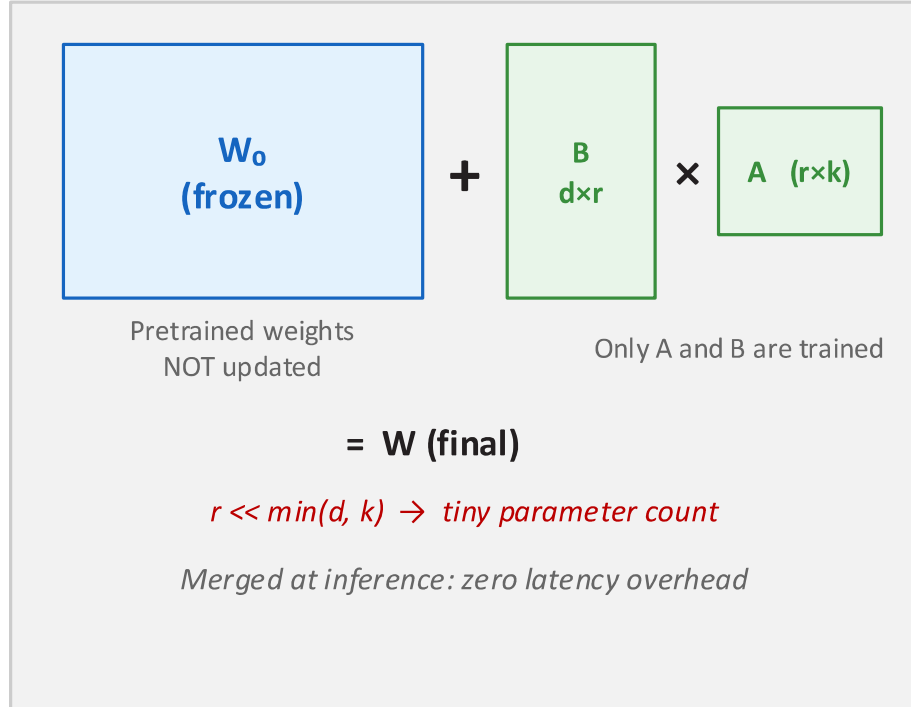
Research Objectives

This thesis investigates five questions about quality-stratified fine-tuning:

- 1 Does quality-stratified rank allocation outperform uniform-rank fine-tuning?**
Compare QS-LoRA against uniform LoRA (B3) and AdaLoRA (B5) across 10 backbones, 3 datasets.
- 2 Does allocating rank by quality beat simply discarding low-quality data?**
Test stratify-and-retain against hard filtering, the dominant existing approach.
- 3 At what model scale does quality-stratified adaptation stop helping, and why?**
Identify the scale boundary and explain it mechanistically via adapter weight geometry.
- 4 Can a distilled local judge replace an API-based judge for quality scoring?**
Evaluate an air-gapped Llama-3.2-3B judge against Claude Haiku for tier agreement.
- 5 Does quality stratification reduce hallucination, not just raw accuracy?**
Measure Unsupported-Answer Rate alongside Gold Alignment across all backbones.

What is LoRA? Low-Rank Adaptation

Instead of updating all billions of model parameters, LoRA adds two small trainable matrices whose product approximates the weight update.



The Rank r controls capacity

- $r = 32 \rightarrow$ high capacity, larger update, more parameters
- $r = 16 \rightarrow$ moderate capacity
- $r = 8 \rightarrow$ low capacity, small update, few parameters

MOTIVATION

Fine-Tuning Cost Across Model Scales

Actual training times on AWS-equivalent A100-SXM4-80GB hardware (QS-LoRA, RepLiQA)

Model	Parameters	GPUs	Wall Time	AWS Cost
Llama-3.2-1B	1B	1	1h 28m	\$7.54
Llama-3.2-3B	3B	1	2h 38m	\$13.45
Llama-3.1-8B	8B	1	4h 07m	\$21.04
Qwen2.5-14B	14B	1	8h 54m	\$45.52
Gemma-3-12B	12B	1	10h 26m	\$53.39
Llama-3.1-70B	70B	4	32h 40m	\$669.07

50× cost difference between 3B (\$13) and 70B (\$669). 70B requires 4 GPUs for 33 hours. The 1B–8B range is the practical deployment target for most organizations.

Fine-Tuning Small Models on Private Documents

01

The Dataset Gap

Private documents lack human-annotated QA datasets, creating an immediate barrier to traditional supervised training.

02

LLM Generation

The only scalable option is to generate synthetic question-answer pairs from the private documents using an LLM.

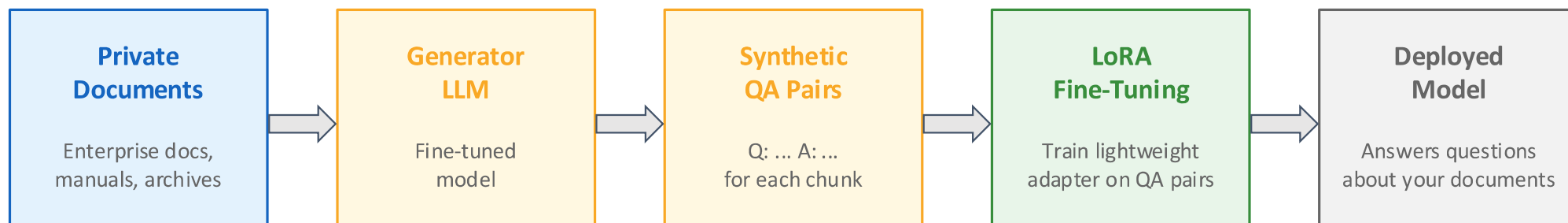
03

Targeted Tuning

Feed those synthetic pairs to LoRA fine-tuning to produce an optimized, domain-adapted model.

Fine-Tuning Small Models on Private Documents

Private document collections have no human-annotated QA datasets. The only scalable path is synthetic data generation.



- No human annotation needed, the generator LLM reads each document chunk and produces question-answer pairs automatically
- The LoRA adapter is trained on these synthetic pairs, it learns to answer questions grounded in your specific documents
- The key bottleneck: synthetic generation produces pairs of highly variable quality, and how we handle that quality variation determines the final model's performance

The Quality Problem in Synthetic Training Data

High Quality

- Grounded in document content
- Requires document to answer
- Well-formed question and answer
- Genuine comprehension needed

Medium Quality

- Mostly grounded, minor gaps
- Some general knowledge overlap
- Usable but not all dimensions strong
- One weak rubric dimension

Low / Drop

- Answer contradicts document
- Trivially answerable without doc
- Malformed or empty answer
- Pure general knowledge test

Yue et al. (2022) showed that 30–54% of synthetically generated QA pairs must be discarded.

Existing Approaches: Filter or Treat Equally

Approach 1: Hard Filtering

- Score each pair → discard below threshold
- Used by: LIMA, AlpaGasus, RAGAS filtering
- Problem: discards real domain signal
- Problem: reduces already-scarce training data

Approach 2: Uniform LoRA (B3)

- Train one adapter on all non-drop pairs at $r=16$
- All pairs receive equal model capacity
- Problem: high-quality pairs get same rank as low-quality pairs
- Problem: noisy signal dilutes the learning

The Gap

No existing method allocates greater model capacity to higher-quality pairs without discarding the lower-quality ones entirely. Quality is applied either as a hard binary filter or ignored completely.

Positioning QS-LoRA in the PEFT Landscape

Parameter-Efficient Fine-Tuning Methods

- QLoRA: 4-bit quantized base + LoRA adapters
- AdaLoRA: adaptively prunes rank per layer during training
- DoRA: decomposes weight updates into magnitude + direction; improves expressiveness at fixed rank

Closest Prior Work: TsqLoRA

- **Quality shapes sampling, not capacity** - higher-quality pairs are sampled more often
- **Rank still varies per layer, not per data tier** - same axis as AdaLoRA
- **QS-LoRA varies capacity by data tier instead** - separate adapters per tier

Where QS-LoRA Fits

No prior method combines full per-tier adapter separation with a tier-weighted dense merge, capacity is allocated by data quality, not by layer sensitivity or sampling frequency.

End-to-End Pipeline

1 Generate

3B model writes QA pairs

2 Score

Haiku rubric, 4 dims

3 Stratify

High / Med / Low / Drop

4 Train ×3

$r = 32 / 16 / 8$

5 Merge

$0.6H + 0.3M + 0.1L$

6 Deploy

single dense checkpoint

1

Synthetic QA Generation

Fine-tuned Llama-3.2-3B generates one pair per document chunk

2

Quality Scoring

Claude Haiku scores each pair on 4 rubric dimensions (1–5)

3

Tier Assignment

Drop / Low / Medium / High based on score rules

4

Tier-Specific Training

Three LoRA adapters at $r=32$, $r=16$, $r=8$

5

Dense Merge

Tier-weighted merge ($0.6 / 0.3 / 0.1$) → single checkpoint

6

Deployment

No adapter overhead at inference time

METHOD

Quality Scoring: Four-Dimension Rubric

Claude Haiku 4.5 (AWS Bedrock) scores each synthetic pair in a single API call, all four dimensions returned as JSON.

Grounding	Relevance	Doc Necessity	Overall
Is the answer directly supported by the context?	Does the answer address the question?	Is this document required to answer?	Is this a valuable training example?
5 = every claim in context 3 = mostly supported 1 = contradicts context	5 = directly answers 3 = partially answers 1 = off-topic	5 = impossible without doc 3 = doc helps 1 = general knowledge	5 = high-quality pair 3 = acceptable 1 = poor or misleading

Tier rules: **Drop** if grounding ≤ 2 or empty answer | **Low** if any dimension ≤ 2 | **High** if all ≥ 4 | **Medium** otherwise

QS-LoRA: Allocate Rank Proportional to Quality

Key Idea

Do not discard low-quality pairs and do not treat all pairs equally. Instead, assign each pair to a quality tier and train a separate LoRA adapter per tier, higher quality gets higher rank.

High Quality

$r = 32$

Well-grounded, document-specific,
genuine comprehension required.
All four rubric dimensions ≥ 4 .

Medium Quality

$r = 16$

Usable but imperfect. One or more
dimensions score 3 rather than 4.
Still grounded in document.

Low Quality

$r = 8$

At least one weak dimension (≤ 2).
Retained, still carries domain
vocabulary and lexical signal.

Three adapters merged offline → Single dense checkpoint (weights: 0.6 / 0.3 / 0.1)

Tier Distribution After Quality Scoring

Dataset	Total Judged	High	Medium	Low	Drop	Drop Rate
RepLiQA	13,770	7,668	1,274	643	4,139	30%
Quoref	7,118	2,532	372	364	3,843	54%
SQuAD v2	39,787	20,410	1,870	2,702	14,765	37%

Key observations:

- 30–54% of generated pairs are dropped entirely, confirming substantial quality heterogeneity in synthetic pipelines
- B3 and B5 train on **ALL** generated pairs including these dropped ones, QS-LoRA selectively filters and stratifies
- High-quality tier dominates on all three datasets (55–68% of judged pairs)

Tier-Specific Training and Dense Merge

Training Configuration

LoRA Target Modules:	q, k, v, o, gate, up, down (all 7)
Optimizer:	AdamW, lr = 2e-4
Effective Batch Size:	8 (1 × grad accum 8)
Epochs:	3
Sequence Length:	4,096 tokens
Precision:	bfloat16
Seed:	42 (single run)
Hardware:	1× A100-SXM4-80GB
70B exception:	4-bit QLoRA, 4× A100

Tier-Weighted Dense Merge

$$\begin{aligned}\Delta W_{\text{merged}} = & 0.6 \cdot \Delta W_{\text{high}} \\ & + 0.3 \cdot \Delta W_{\text{medium}} \\ & + 0.1 \cdot \Delta W_{\text{low}}\end{aligned}$$

Experimental Setup

Evaluation Datasets

RepLiQA
2,000 questions

Private enterprise documents,
unanswerable questions

Quoref
2,418 questions

Co-reference resolution, Wikipedia
passages

SQuAD v2
11,873 questions

Unanswerable + extractive, Wikipedia
passages

Baseline Conditions

B3

Uniform LoRA $r=16$ on all generated pairs

B5

AdaLoRA on all generated pairs

Ours

QS-LoRA: $r=32/16/8$ per tier, merged dense

Evaluation Metric & Backbone Models

Evaluation Metric

**Gold Alignment
(GA,1-5)** Scored by Claude Haiku judge

**Unsupported -
Answer Rate** Fraction of answered question asserting
unsupported content

**Grounding &
Relevance** Is the answer supported by context?

Backbone Models

Llama-3 1B / 3B / 8B / 70B

Qwen-2.5 3B / 7B / 14B

Gemma-3 1B / 4B / 12B

RESULTS

Results: Small Models (1B–8B)

QS-LoRA outperforms B3 on all 7 small models across all 3 datasets. 95% CI shown for Llama-3.2-3B (reference).

RepliQA

Model	B3	Ours	Δ
Llama-1B	3.38	3.53	+0.15
Llama-3B	3.64	3.78	+0.14
Llama-8B	3.66	3.80	+0.14
Qwen-3B	3.46	3.60	+0.15
Qwen-7B	3.59	3.74	+0.15
Gemma-1B	3.00	3.19	+0.20
Gemma-4B	3.43	3.65	+0.22

Quoref

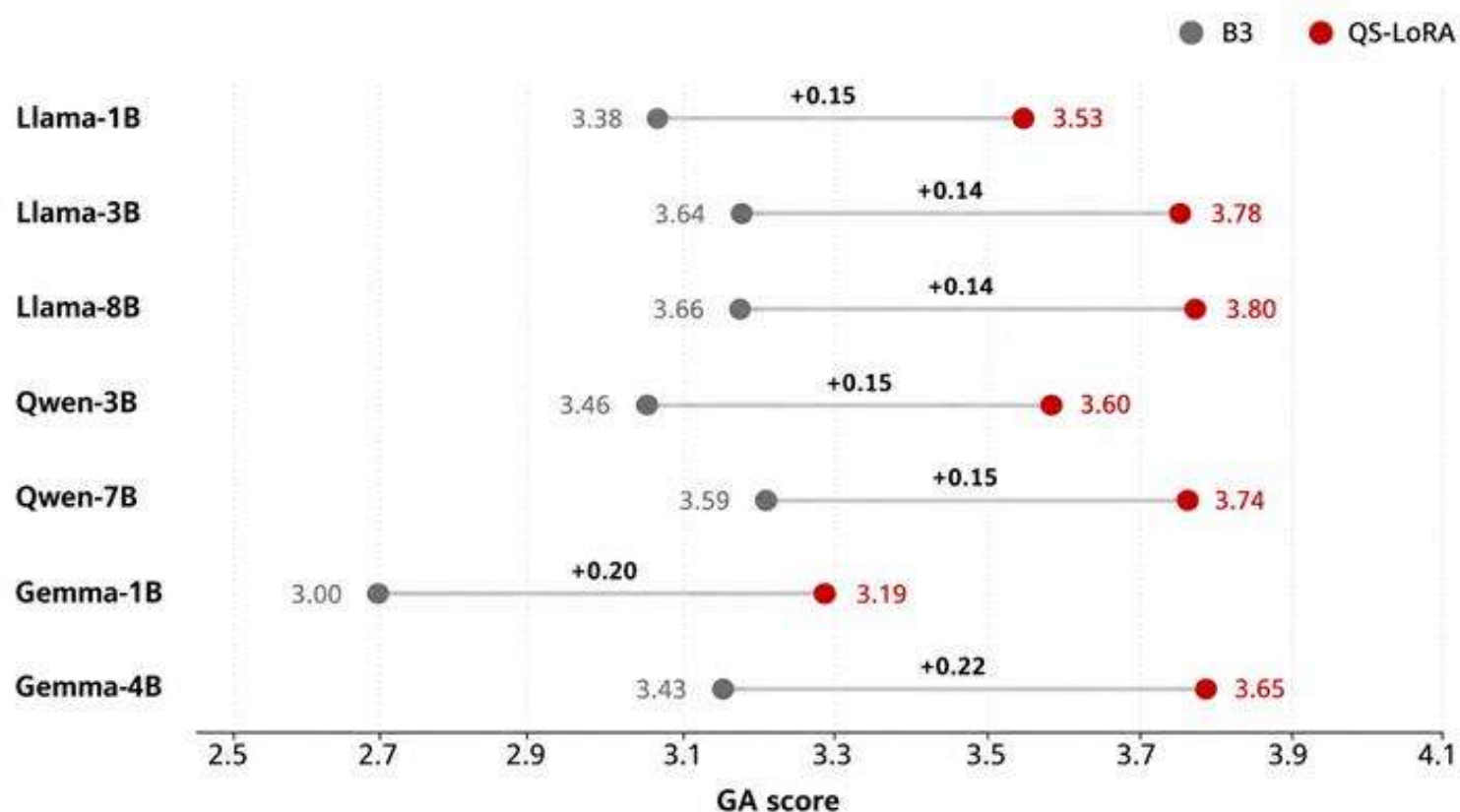
Model	B3	Ours	Δ
Llama-1B	2.69	2.82	+0.13
Llama-3B	3.50	3.71	+0.20
Llama-8B	3.87	4.03	+0.16
Qwen-3B	3.56	3.71	+0.15
Qwen-7B	3.75	3.88	+0.13
Gemma-1B	2.59	2.72	+0.14
Gemma-4B	3.59	3.82	+0.23

SQuAD v2

Model	B3	Ours	Δ
Llama-1B	1.90	2.03	+0.14
Llama-3B	2.16	2.28	+0.11
Llama-8B	2.17	2.31	+0.14
Qwen-3B	2.17	2.30	+0.13
Qwen-7B	2.20	2.33	+0.13
Gemma-1B	1.81	1.93	+0.12
Gemma-4B	2.11	2.22	+0.11

RESULTS

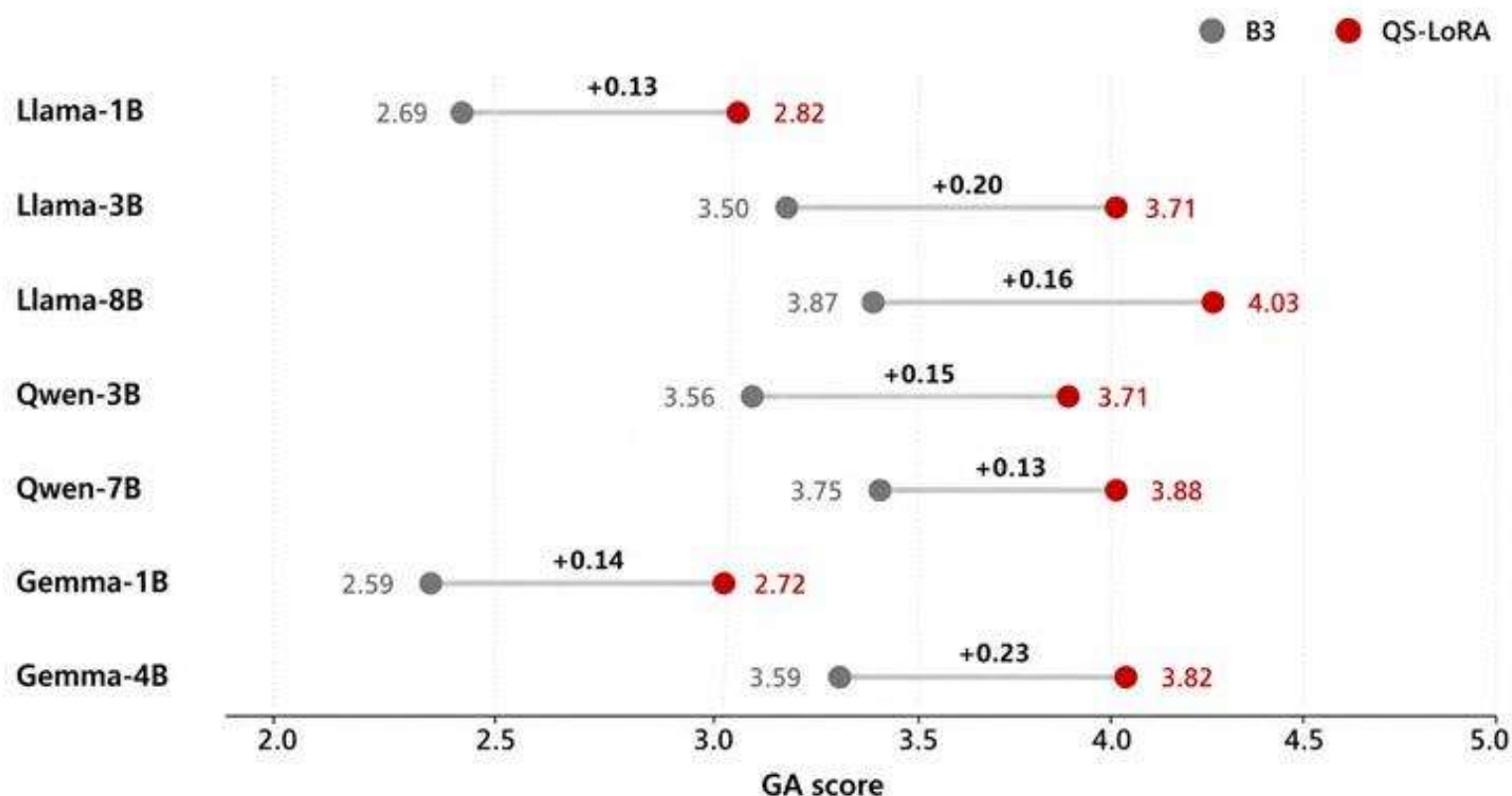
Results: Small Models (1B–8B) - RepLiQA



QS-LoRA wins on all 7 models, gain ranging from +0.14 to +0.22 GA, largest on Gemma-4B (+0.22).

RESULTS

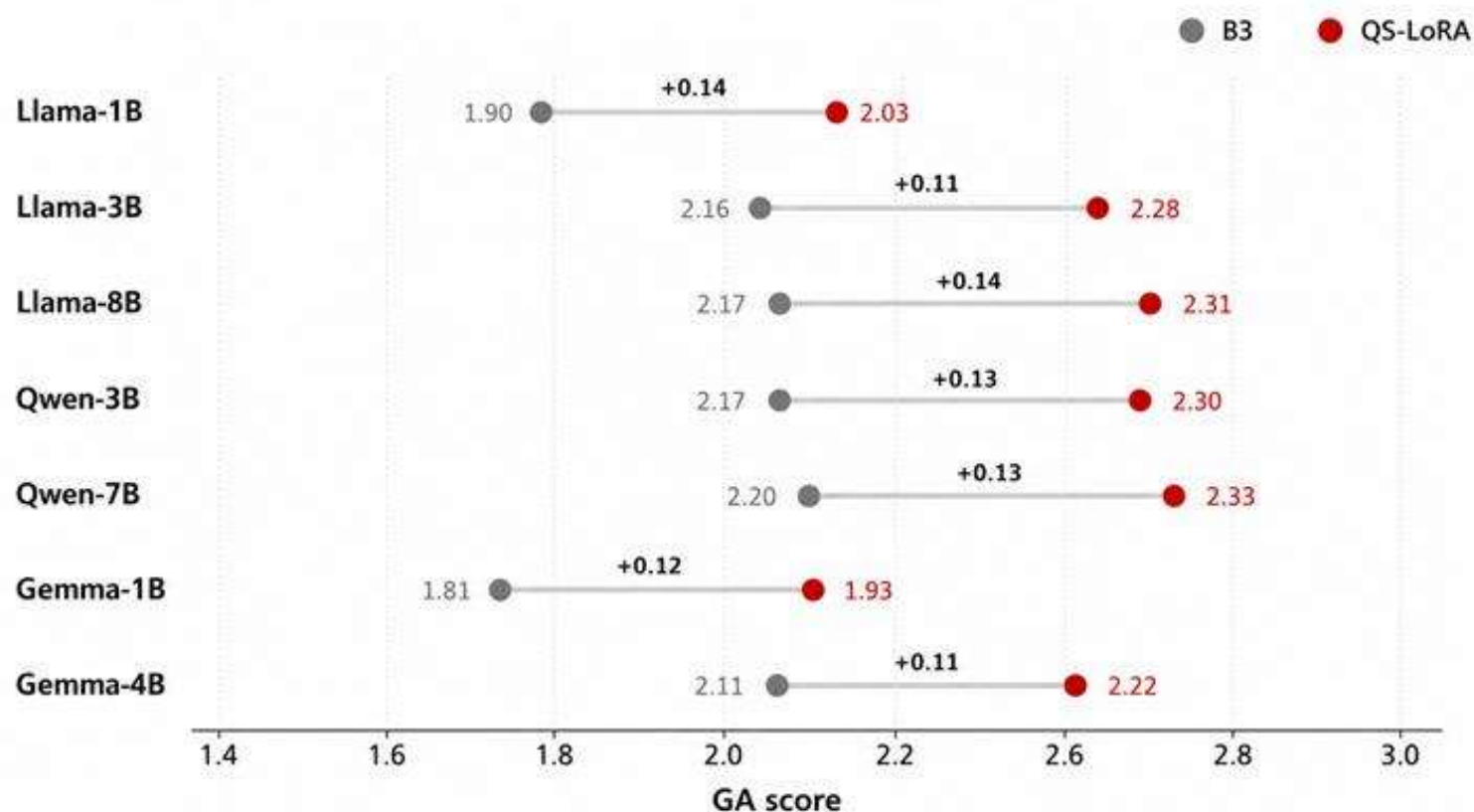
Results: Small Models (1B–8B) - Quoref



QS-LoRA wins on all 7 models, gain ranging from +0.13 to +0.23 GA, largest on Gemma-4B (+0.23).

RESULTS

Results: Small Models (1B–8B) - SQuAD v2



QS-LoRA wins on all 7 models, gain ranging from +0.11 to +0.14 GA, largest on Llama-8B (+0.14).

RESULTS

Results: Large Models (12B–70B) and Scale Boundary

RepLiQA

Model	B3	Ours	B5
Qwen2.5-14B	3.70	3.86	3.96
Gemma-3-12B	3.67	3.77	3.89
Llama-3.1-70B	3.76	3.87	4.10

Quoref

Model	B3	Ours	B5
Qwen2.5-14B	4.09	4.20	4.28
Gemma-3-12B	3.74	3.93	4.12
Llama-3.1-70B	4.10	4.16	4.35

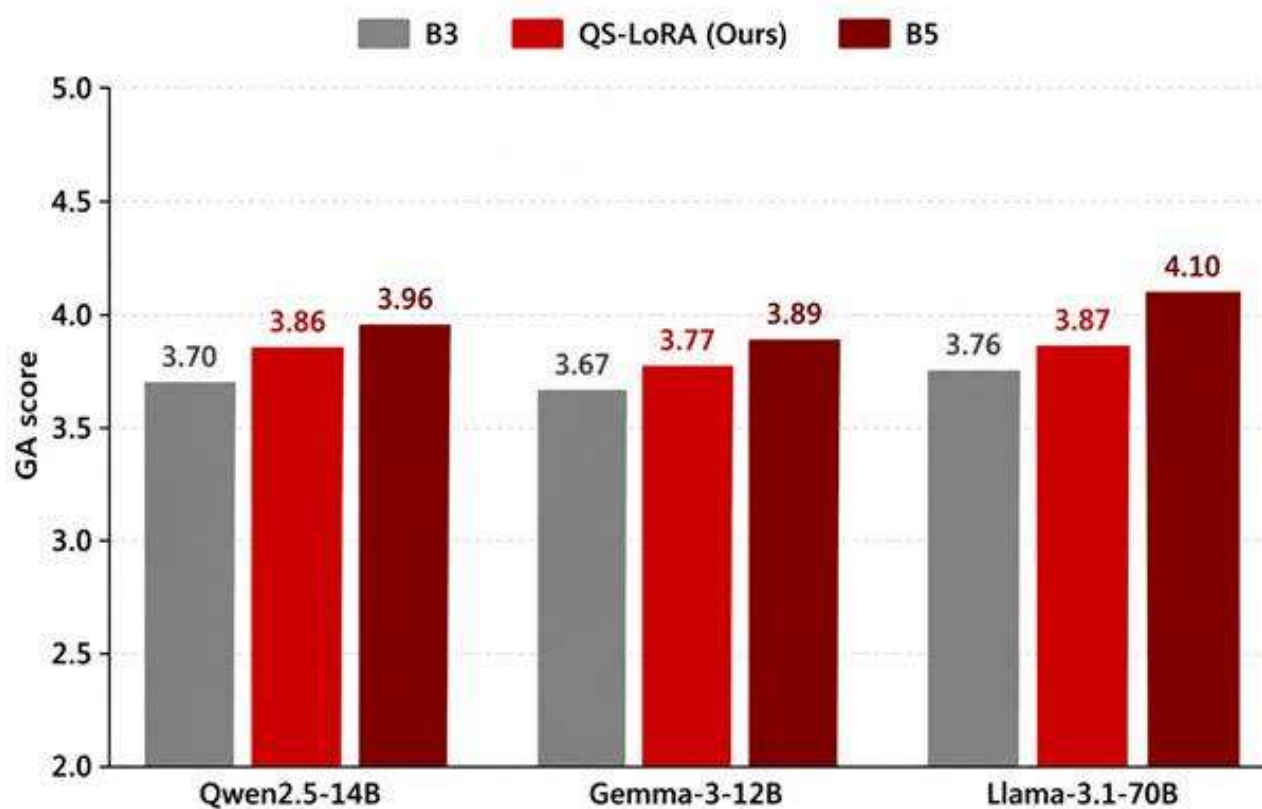
SQuAD v2

Model	B3	Ours	B5
Qwen2.5-14B	2.27	2.41	2.75
Gemma-3-12B	2.32	2.40	2.55

QS-LoRA beats B3 in all 8 large-model calls; AdaLoRA (B5) beats QS-LoRA in all 8

RESULTS

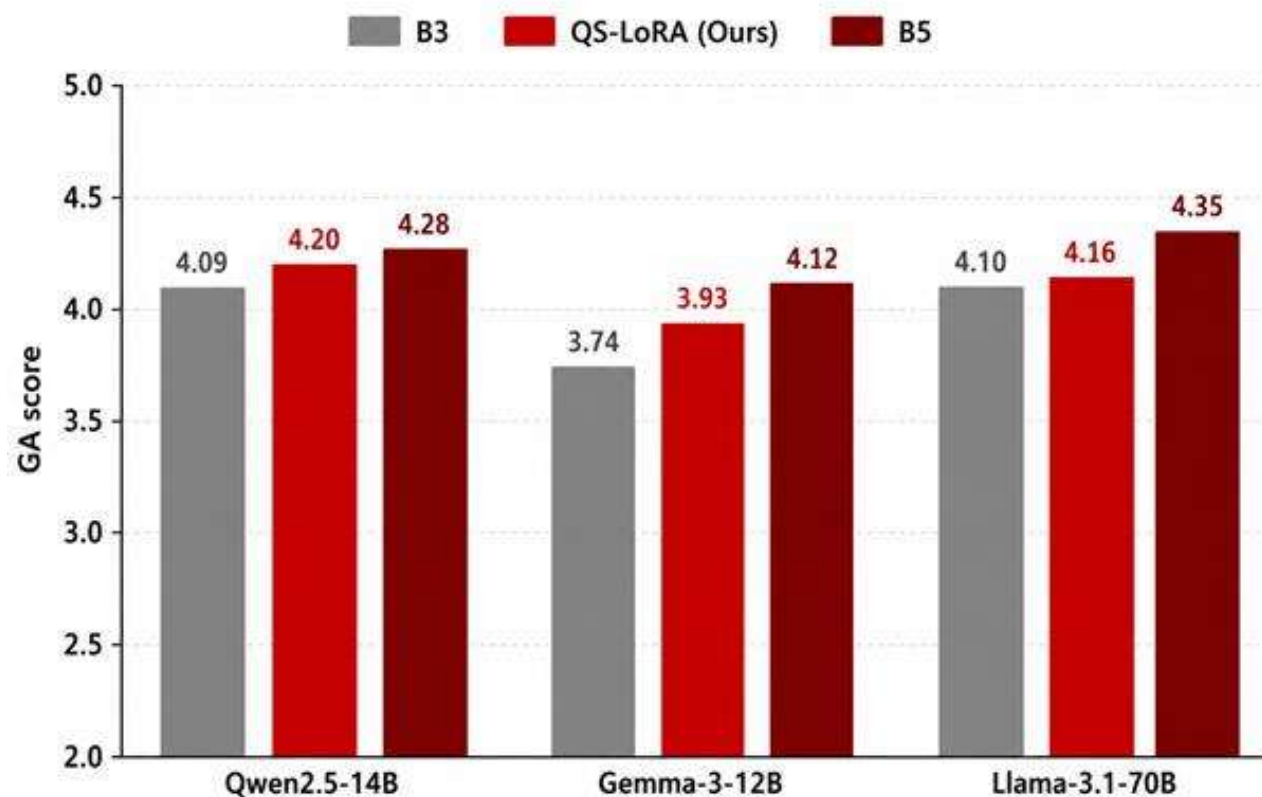
Results: Large Models (12B–70B) - RepLiQA



AdaLoRA (B5) wins here too, the margin widens with scale, from +0.10 at Qwen-14B to +0.23 at Llama-70B

RESULTS

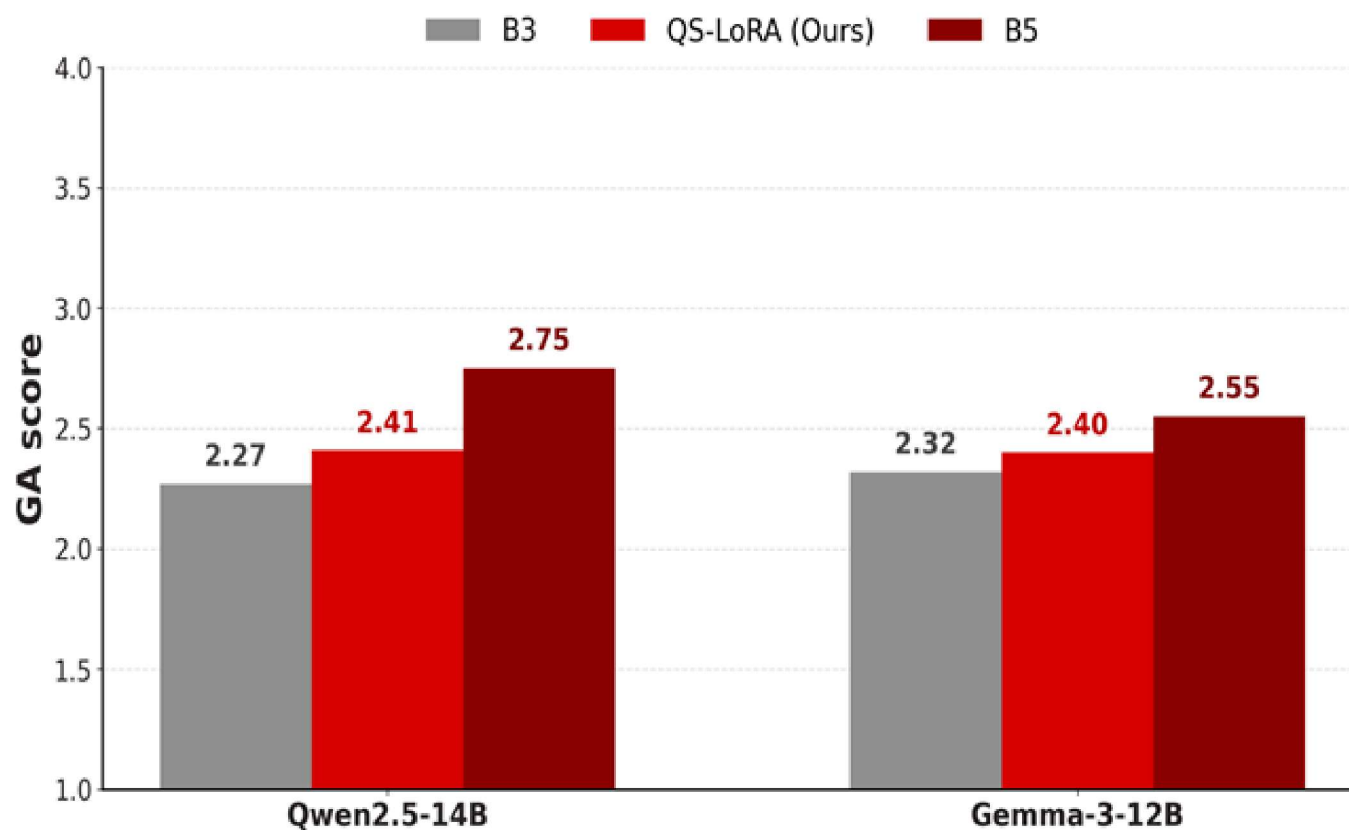
Results: Large Models (12B–70B) - Quoref



Widest AdaLoRA margins on Quoref, +0.19 GA at both Gemma-12B and Llama-70B.

RESULTS

Results: Large Models (12B–70B) - SQuAD v2



Steepest
AdaLoRA
advantage in the
entire study,
Qwen-14B trails
B5 by +0.34 GA

Why AdaLoRA Wins at Large Scale: Weight Geometry

SVD analysis of actual adapter weight update matrices ($\Delta W = B \cdot A$) across five backbone models — RepLiQA adapters.

Mean Frobenius Norm $\|\Delta W\|_F$ per module

Model	B3	Ours	B5	B5/Ours ratio
Llama-3.2-3B	4.83	2.85	0.76	0.267
Llama-3.1-8B	5.61	3.35	0.61	0.182
Qwen2.5-3B	4.16	2.42	0.53	0.219
Qwen2.5-14B	5.42	3.49	0.25	0.071
Llama-3.1-70B	8.47	5.43	0.20	0.036 ←

At 70B: AdaLoRA Frobenius = 0.20 vs QS-LoRA = 5.43 — a 27× difference. AdaLoRA wins by making smaller, more precise updates, not larger ones.

Weight Geometry: Three Findings

01

Smaller Updates

AdaLoRA makes **3–27× smaller updates** than QS-LoRA, and shrinks as scale grows (0.76 at 3B → 0.20 at 70B).

02

Highest Effective Rank

QS-LoRA has highest effective rank (~29 vs ~13 for baselines) — indicating that energy is spread across many directions.

03

Targeted Corrections

Large models need only **tiny targeted corrections**. QS-LoRA's broad updates disrupt already-correct representations.

RESULTS

How Close to Paid API Models?

Ceiling references: Claude Opus 4.8 (GA = 4.40 on RepLiQA, 4.45 on Quoref).

RepLiQA

Model	Ours GA	Gap to Opus	% Gap closed
Llama-3.2-1B	3.53	0.87	14.7%
Gemma-3-1B	3.19	1.21	13.6%
Llama-3.2-3B	3.78	0.62	18.4%
Llama-3.1-8B	3.80	0.60	18.9%
Qwen2.5-14B	3.86	0.54	22.9%
Llama-3.1-70B	3.87	0.53	17.2%

Quoref

Model	Ours GA	Gap	% Gap closed
Llama-3.2-1B	2.82	1.63	7.4%
Gemma-3-1B	2.72	1.73	7%
Llama-3.2-3B	3.71	0.74	22.1%
Llama-3.1-8B	4.03	0.42	27.6%
Qwen2.5-14B	4.20	0.25	30.6%
Llama-3.1-70B	4.16	0.29	17.1%

- Qwen2.5-14B on Quoref closes 30.6% of the B3-to-Opus gap (GA 4.20 vs ceiling 4.45)
- Gap-closed rises steadily from 1B through Qwen2.5-14B on both datasets, then narrows at 70B.
- Even modest fine-tuning runs meaningfully close the gap to a paid API model.

RESULTS

Hallucination Analysis: Unsupported-Answer Rate

UAR = fraction of answered questions where model asserts content not supported by the source document (grounding score ≤ 2). Lower is better.

UAR (%) on RepLiQA — Ours beats B3 on all 10 models

Model	B3	Ours	Δ vs B3
Llama-3.2-1B	34.2	26.5	-7.7
Llama-3.2-3B	31.7	22.0	-9.7
Llama-3.1-8B	29.6	20.4	-9.2
Llama-3.1-70B	25.8	15.8	-10
Qwen2.5-3B	35.6	25.8	-9.8
Qwen2.5-7B	32.2	21.7	-10.5
Qwen2.5-14B	29.9	17.3	-12.6
Gemma-3-1B	46.0	36.4	-9.6
Gemma-3-4B	33.3	23.2	-10.1
Gemma-3-12B	29.7	18.5	-11.2

RESULTS

Hallucination Analysis: Unsupported-Answer Rate

93.5%

Coverage

vs ~100% for B3

31.6%

Correct-Refusal

on unanswerable questions

~0%

Over-Refusal

on answerable questions

Emergence from Quality Stratification: Not designed, it emerged from quality stratification: the high-quality tier's r=32 adapter learned to distinguish answerable from unanswerable questions more sharply than baselines trained on mixed-quality data at uniform capacity.

RESULTS

Hallucination Examples: B3 Invents, QS-LoRA Refuses

REFUSAL WIN

Llama-3.2-3B / RepLiQA

Q: What innovative waste management practices emerged post-2023 clean-up events?

B3: The events provided a platform... One such practice was the launch of a smartphone app that rewarded users for recycling, introduced on November...

Ours: None. The context does not mention any innovative waste management practices.

REFUSAL WIN

Qwen2.5-14B / SQuAD v2

Q: When was Garrison born?

B3: Fielding H. Garrison was born in 1862.

Ours: The text does not provide information about Garrison's birth date.

ANSWER WIN

Llama-3.2-3B / Quoref

Q: Last name of writer who wrote that the woman 'would have drawn whistles of astonishment in Bedlam'?

B3: Love (confuses subject of quote with its author)

Ours: Peel

In refusal wins: B3 invents specific facts (a date, a year) not in the document. In answer wins: B3 extracts the wrong entity. QS-LoRA's high-quality tier training gives stronger document grounding.

Ablation Studies: Does the Merge Weighting Scheme Matter?

Tier vs Equal Weight Merge

Scheme	Weights(H/M/L)	RepLiQA GA	Quoref GA	SQuAD v2
Tier-weighted (Ours)	0.60/0.30/0.10	3.78	3.74	2.32
Equal weights	0.33/0.33/0.33	2.39	2.21	1.78

Same three trained adapters (r=32/16/8), only the merge coefficients differ

Key Finding

- Equal-weight merging collapses far below B3 on all three datasets.
- Letting the low-tier adapter's noisy updates cancel the high-tier adapter's signal degrades outputs into repetition loops.
- Quality-aware downweighting of the low adapter is essential.

Ablation Studies: Does the Equal-Weight Collapse Hold Across Models?

RepLiQA GA (Tier vs. Equal Weights)

Scheme	Tier (Ours)	Equal Weights
Llama-3.2-1B	3.69	2.29
Llama-3.2-3B	3.78	2.39
Llama-3.1-8B	3.85	2.31
Qwen-2.5-3B	3.80	2.02
Qwen-2.5-7B	3.92	2.26
Qwen-2.5-14B	3.94	2.30
Gemma-3-1B	3.37	1.32
Gemma-3-4B	3.74	1.55
Gemma-3-12B	3.87	1.59

- The collapse holds on all 9 backbones without exception, not an artifact of one model.
- Tier-weighting is best or tied-best on all 26 complete cross-model cells across RepLiQA, Quoref, and SQuAD v2.

Ablation: Do Low-Quality Pairs Actually Help?

Same tier-weighted merge, only the low-tier adapter's merge weight is zeroed out

Scheme	Weights (H/M/L)	RepLiQA	Quoref	SQuAD v2
High+medium only	0.67/0.33/0.00	3.66	3.60	2.18
Tier-weighted (Ours)	0.60/0.30/0.10	3.78	3.74	2.32

Key Finding

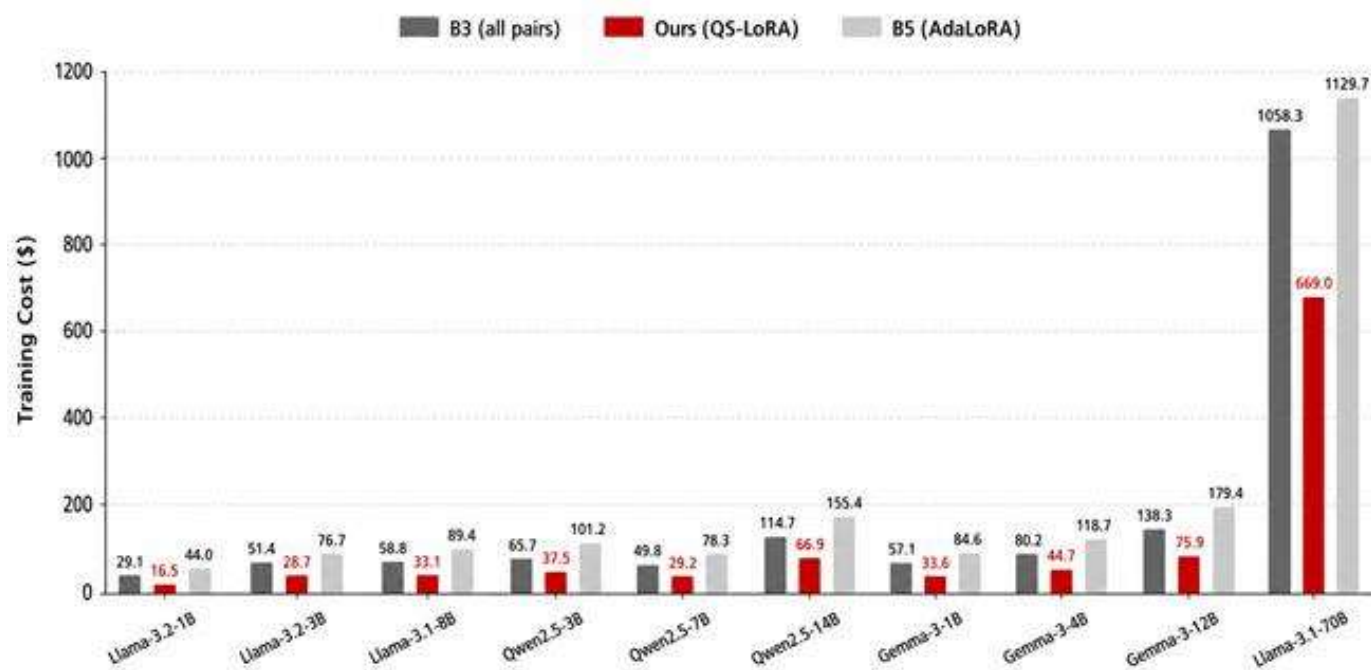
- Dropping the low-tier adapter's contribution costs 0.12–0.14 GA on every dataset.
- Even the lowest-quality tier contributes recoverable signal, consistent with the thesis's broader finding that hard filtering discards value QS-LoRA's stratified allocation preserves.

RESULTS

Training Cost: QS-LoRA vs Baselines

B3 and B5 costs scaled to train on ALL generated pairs (no filtering). QS-LoRA trains on non-drop pairs only. AWS rate: \$5.12/hr per A100-80GB.

RepLiQA training cost (\$) per model



Does a Distilled Local Judge Replace the API Judge?

Distilled Judge vs Claude Haiku

Domain	n	Agreement
RepLiQA	1,253	82.60%
SQuAD v2	3,830	81.25%
OhioLine (out-of-domain)	511	79.85%
Quoref	671	80.50%
DROP (zero-shot)	1,045	84.50%
Pooled	7,310	81.72%

Key Finding

- The distilled judge agrees with Haiku on 79.85–84.50% of pairs across all five domains
- OhioLine is a fully out-of-domain agricultural corpus never seen in training, and DROP is zero-shot, both score in the same range as in-domain results.

Limitations

01

Scale Boundary

QS-LoRA is not recommended at 12B+. AdaLoRA's conservative updates are better calibrated for large pretrained models.

02

Single-Seed Training

All conditions trained once (seed=42); bootstrap CIs quantify evaluation uncertainty but not cross-seed variance.

03

No Human Evaluation

GA is scored by the same judge (Haiku) used to score training data; gains may partly reflect judge-preference alignment.

Future Work

01

Scale-Adaptive Rank

Make tier ranks scale-dependent so larger models get smaller per-tier ranks.

02

Combining with AdaLoRA

QS-LoRA allocates rank on data axis, AdaLoRA on parameter axis; these can be combined.

03

Instance-Weighted Loss

A simpler single-adapter alternative that upweights high-quality pairs in the loss.

04

Human Evaluation

Validate GA judge scores against domain-expert ratings (50–200 responses) to confirm quality.

Contributions

01

QS-LoRA Method

Outperforms B3 in all 29 cells, B5 in 21/29.

02

End-to-End Pipeline

No human annotation, fully on-premise.

03

10-Model Study

Scale boundary mechanistically explained by SVD analysis.

04

Distilled Local Judge

81.72% pooled tier agreement, 5 domains, air-gapped deployment.

05

Cost & Hallucination

QS-LoRA cheaper than both baselines; reduces UAR on 10/10 models.

Conclusion

1 **Quality stratification consistently improves fine-tuning**

QS-LoRA beats uniform LoRA in all 29 evaluated model-dataset combinations. The benefit is robust across model families, datasets, and scales in the 1B–8B range.

2 **Low-quality pairs carry recoverable signal**

Excluding them (high-only ablation) degrades performance by 0.12–0.13 GA points across all three datasets. Stratification retains this signal at appropriate rank.

3 **The scale boundary is mechanistically understood**

SVD analysis shows QS-LoRA makes 5–27× larger updates than AdaLoRA at 12B–70B, overshooting pretrained representations that need only small corrections.

4 **Quality stratification reduces hallucination**

QS-LoRA reduces unsupported answers vs B3 on all 10 models. On Llama-3.2-3B RepLiQA, UAR drops from 26.7% to 13.7% through learned calibrated abstention.